Using Large Language Models for Rubric-Based Assessment

Madhav Khanal and Dan S. Myers
Department of Computer Science, Rollins College, Winter Park, FL 32789

Introduction

The rapid adoption of Large Language Models(LLMs) in educational settings presents both opportunities and risks for academic assessment-related tasks. Automated grading has become an essential tool in educational technology because of its ability to assess large volumes of student work, provide consistent evaluations and deliver immediate feedback to enhance learning. Some research has shown that LLMs can provide higher reliability in rubric based assessment (Yavuz 2025) while others have questioned its validity (Yeung et al 2025). This research investigates whether leading current generation models can align with human experts on realistic program assessment tasks. We prompted a collection of leading models from OpenAI (GPT-5, GPT-4o, o3 and o4-mini) and Anthropic (Claude Sonnet 4 and Opus 4.1) to score 100 expert graded student artifacts. These artifacts come from a first-year general education seminar at Rollins College and were evaluated using the AACU Information Literacy VALUE rubric(Association of American Colleges & Universities, n.d.). We examined model self-consistency, alignment with expert faculty evaluators and the effect of model choice and prompting techniques on results.

Methodology

Our dataset comes from a sample of 100 anonymised artifacts collected from first-year general education seminar courses at Rollins College during the 2024-2025 academic year. Each artifact was assessed by three faculty evaluators using the AACU Information Literacy VALUE rubric. The rubric has five categories with each category scored on a scale from 0 (lowest) to 4 (highest). The resulting dataset has 1500 observations (3 reviewers * 100 artifacts * 5 categories). Each AI model scored each of the 100 artifacts 3 times through API calls. Prompting techniques can significantly impact LLM scoring accuracy (Kim, 2025). We focused on two prompting strategies:

- Zero-Shot prompting, where the model is provided with an artifact, a copy of the rubric, and a prompt of basic scoring instructions.
- Few-Shots prompting, where the prompt includes example artifacts and expert scores to provide additional guidance. We used a three-shot approach with high, medium and low scoring examples.

Our evaluation focuses on the percentage of artifacts achieving a total score of at least 10 (human evaluators placed 50.7% of artifacts in this category). We also examine the overall distribution of scores across rubric categories.

Results

1.) Model Reliability

- All models showed higher inter rater reliability (ICC .71-.86) compared to human experts (.792)
- However, score distributions didn't align with human patterns

2.) Zero Shot Prompting

- All models failed to match human distributions
- OpenAI models consistently overscored: 79-86% of artifacts scored ≥10 (vs. 50.7% human)

3.) Three Shot Prompting (Fig 2)

- **GP-4o**: Best Performance (57.56% of scores >= 10) and matched symmetric human distribution
- **GPT-5 and o3**: Persistent overscoring (71.5% and 73.7% ≥10), unaffected by prompting strategy
- Claude Sonnet 4 and Opus 4.1: Compressed distribution (>50% of scores at level 2)
- Reasoning models didn't perform better than non-reasoning models. In fact, they struggled with following instructions.

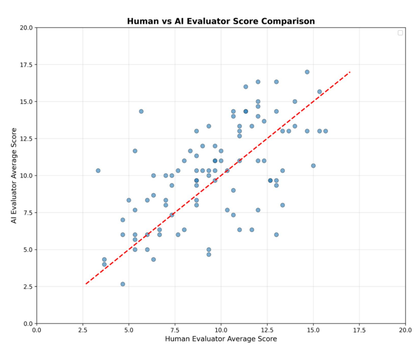

Figure 1: Correlation between AI scores and human experts scores for GPT-4o 3 shots prompt

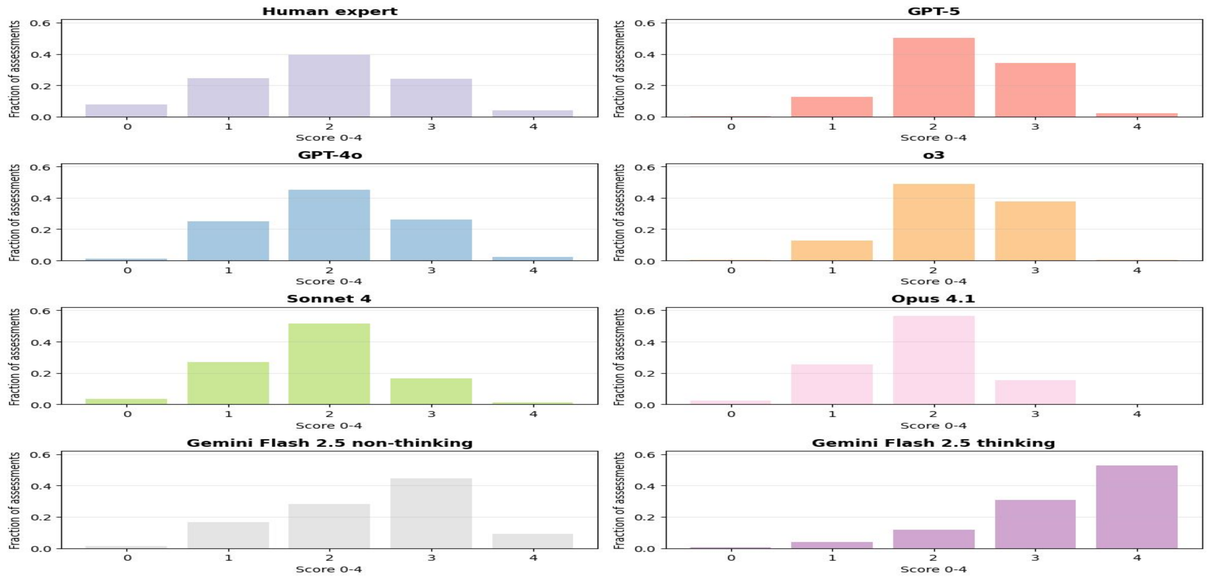

Figure 2: Overall distribution of rubric scores per model for three shot prompting

Conclusions

Key Findings:

- LLMs show high consistency but poor distributional alignment with humans
- Zero shot prompting failed across all models
- GPT-4o with three shots prompting performed best (57.6% vs 50% human)

Implications and Recommendations

- **Use caution**: Model choice and updates significantly affect outcome
- **Limit to low stake**: AI cannot substitute human judgement for high-stakes tasks
- Requires **prompt engineering**: Straightforward prompting is inefficient
- **Establish protocols**: Set benchmarks and retest after model updates
- **Future Direction**: Test simpler tasks (pass/fail) and discipline specific rubrics

References

- Association of American Colleges and Universities. (2024). VALUE Rubrics. https://www.aacu.org/value-rubrics
- Kim, Y. (2025). Automated Essay Scoring With GPT-4 for a Local Placement Test: Investigating Prompting Strategies, Intra-Rater Reliability, and Alignment With Human Scores. TESOL Quarterly.
- Yavuz, F., C ̧elik, O ̈., & Yava ̧s C ̧elik, G. (2025). Utilising large language models for EFL essay grading: An examination of reliability and validity in rubric-based assessments. British Journal of Educational Technology, 56(1), 150–166. https://doi.org/10.1111/bjet.13494
- Yeung, C., et al. (2025). A Zero-Shot LLM Framework for Automatic Assignment Grading in Higher Education. arXiv preprint arXiv:2501.14305.

Acknowledgements

I would like to thank Tony Holbrook and Caitlyn Deter for their assistance in obtaining student artifacts. I also thank Ashley Kister and Lucy Littler for financial support. Additionally, I acknowledge Emmy Frederickson for contributing results for Gemini Flash 2.5.